

Message Directory App

Student Information

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Course: Mobile Programming

Assignment: React Native Message Directory Application

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1. Project Overview

The Message Directory App is a mobile application developed using React Native. The application provides users with a collection of message directories organized into categories. Users can select a directory and view the messages associated with that category.

The application demonstrates the use of React Native components, navigation between screens, data management, and responsive mobile user interface design.

The project including all code, screenshots and readme is also uploaded to github as per the instructions. It is available at <https://github.com/pranayar/MProgramming-EXERCISE-3/>

2. Objectives

The objectives of this project are:

- Develop a mobile application using React Native.
- Implement navigation between multiple screens.
- Display categorized information using a structured interface.
- Create a user-friendly and responsive mobile application.
- Demonstrate Android emulator deployment and testing.

3. Technologies Used

Technology	Purpose
React Native	Mobile application development
JavaScript	Application logic
React Navigation	Screen navigation
Android Studio	Android emulator and testing
Node.js	Runtime environment
npm	Package management

4. Application Features

The application contains the following directories:

1. You
2. Home
3. Love
4. Family
5. Friends
6. School

Functional Features

- Displays six categorized directories.
- Uses color-coded circular directory buttons.
- Supports navigation between screens.
- Displays messages related to the selected category.
- Supports Android emulator execution.
- Responsive layout suitable for mobile devices.

5. Project Structure

MessageDirectoryApp

```
|
|
| └─ src
|   |
|   | └─ data
|   |   └─ messages.js
|   |
|   |
|   | └─ screens
|   |   └─ HomeScreen.js
|   |   └─ MessageScreen.js
|   |
|   |
|   └─ components
|
|
| └─ App.tsx
|
| └─ package.json
└─ README.md
```

File Description

App.tsx

- Entry point of the application.
- Configures navigation between screens.

HomeScreen.js

- Displays all message directories.
- Handles user selection of directories.

MessageScreen.js

- Displays messages associated with the selected directory.

messages.js

- Stores all directory and message data.

6. Installation and Execution

Prerequisites

- Node.js
- npm
- Android Studio
- Android SDK
- Android Emulator

Install Dependencies

npm install

Start Metro Bundler

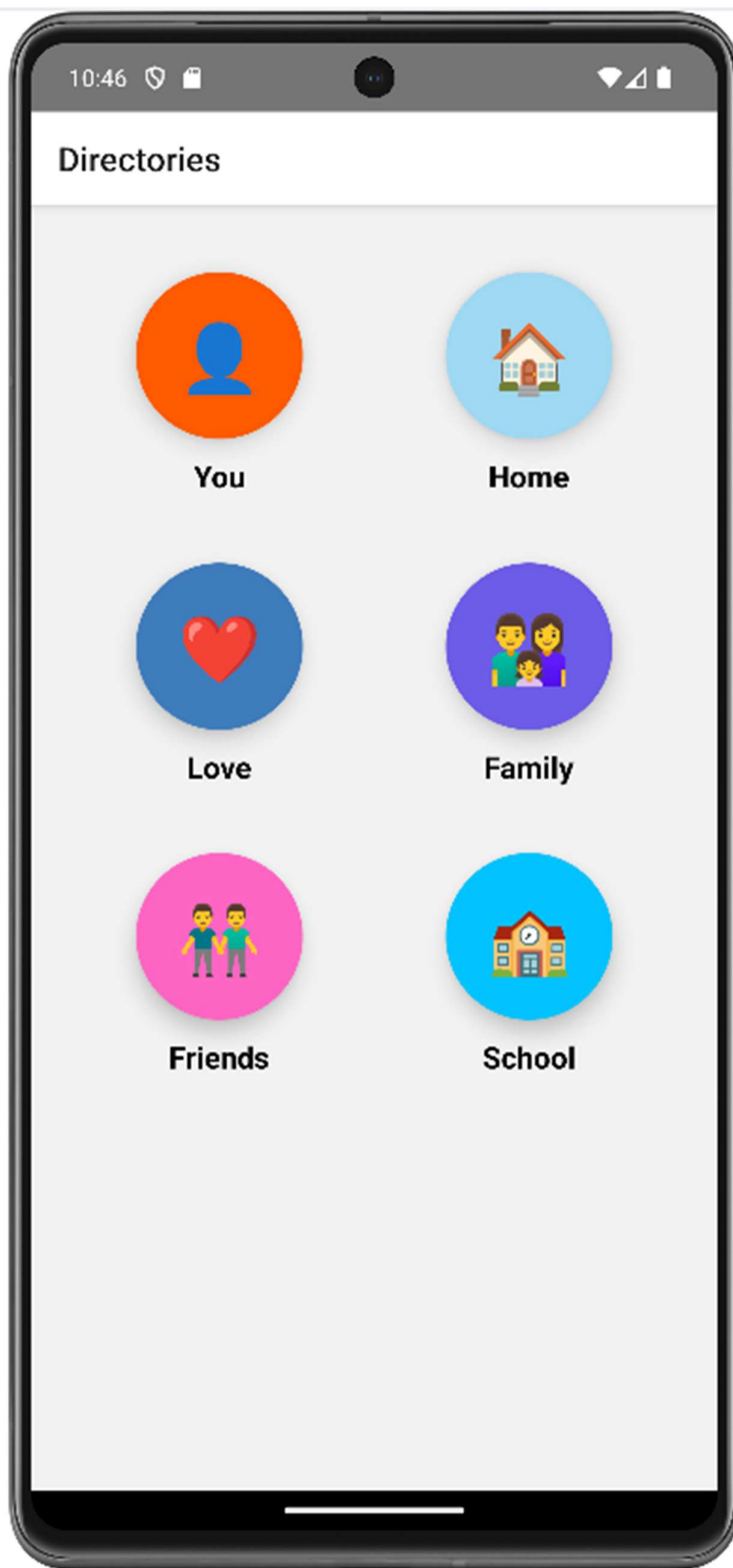
npx react-native start

Run Application

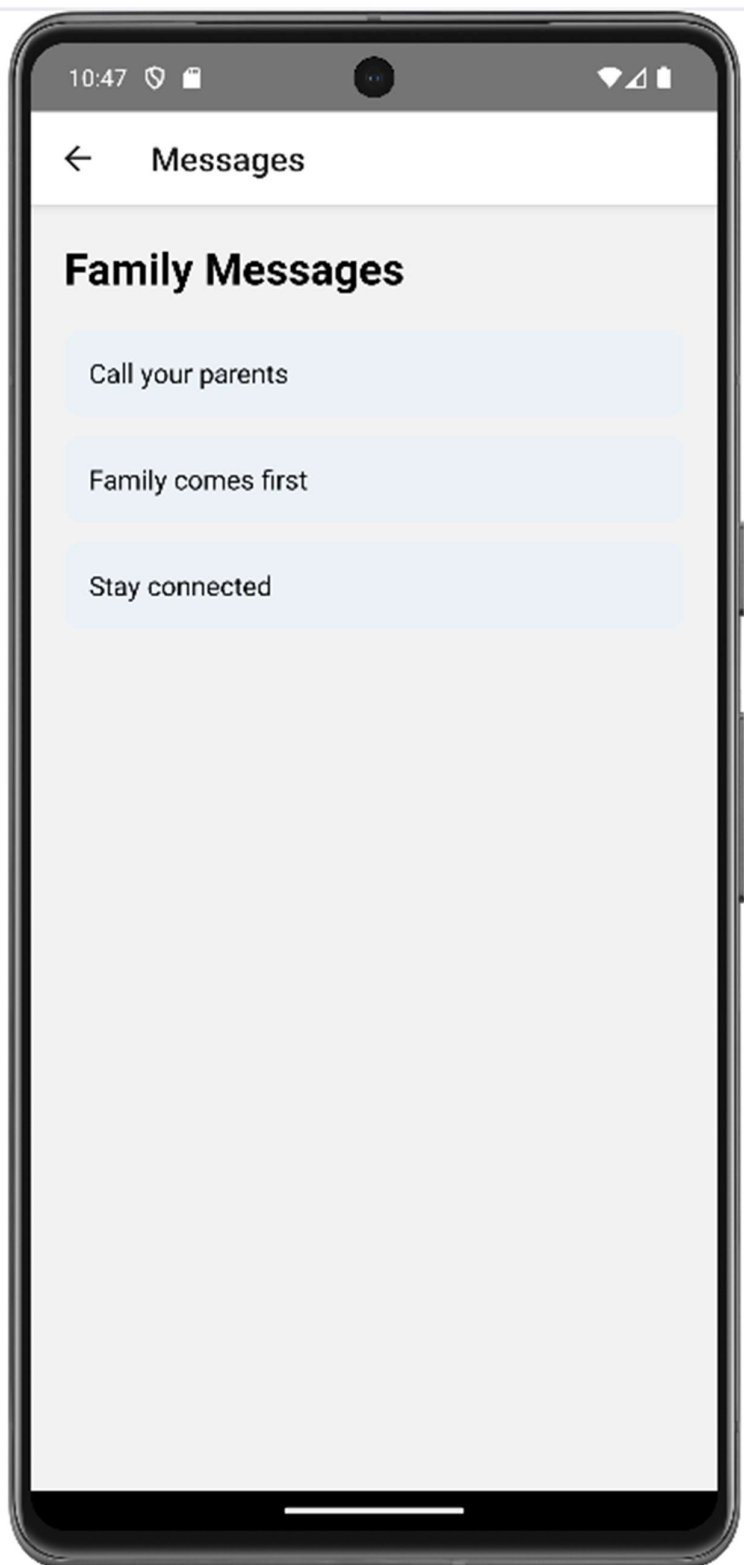
npx react-native run-android

7. Screenshots

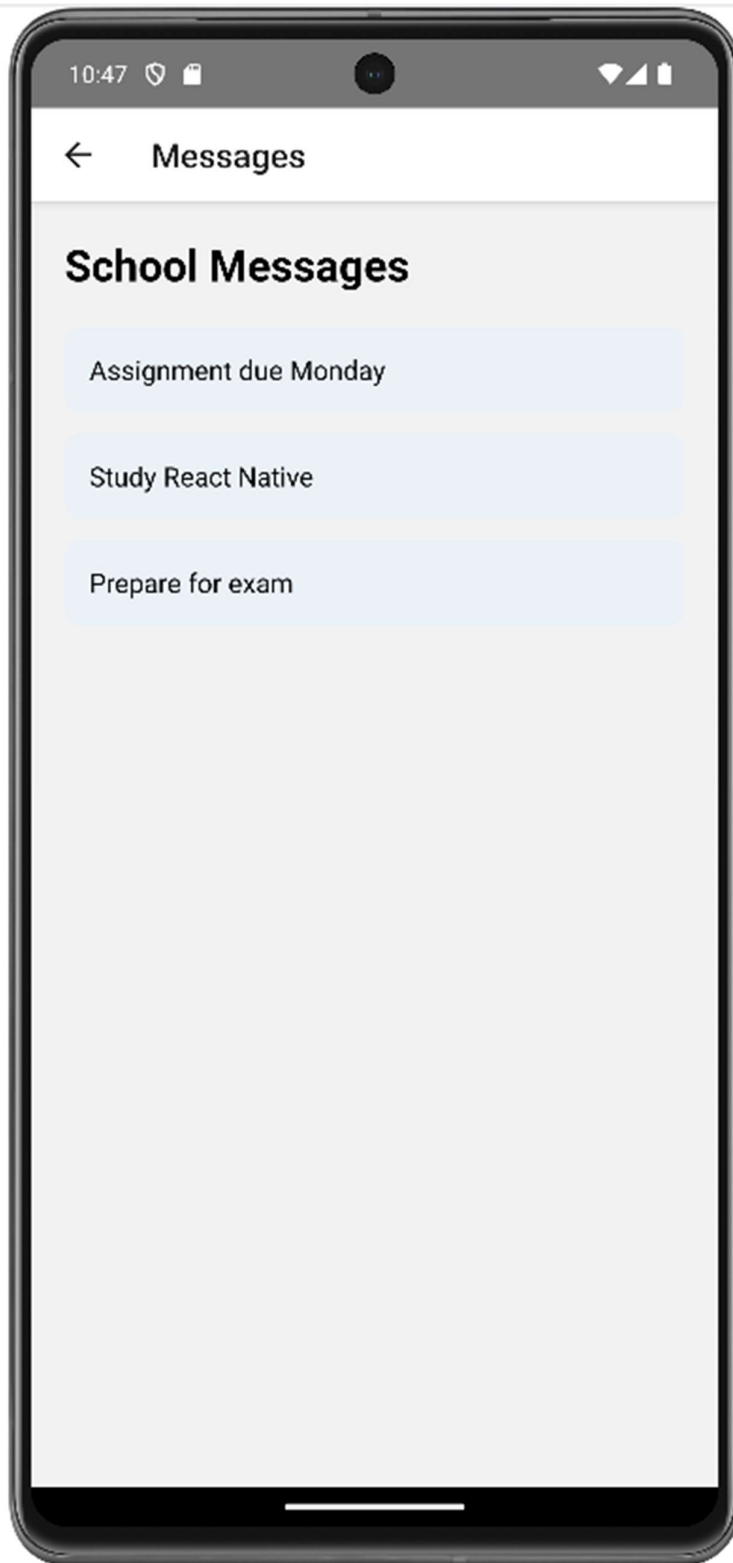
Home Screen



Family Directory Screen



School Directory Screen



8. Challenges Encountered

During development, the following issues were encountered:

- Android SDK configuration issues.

- Environment variable setup for Android SDK.
- Android emulator storage limitations.
- React Native project configuration and navigation setup.

These issues were successfully resolved through proper SDK configuration and emulator management.

9. Conclusion

The Message Directory App successfully meets all assignment requirements. The application provides categorized message directories, navigation between screens, and message display functionality using React Native. The project demonstrates practical experience with mobile application development, Android deployment, and user interface design.

10. References

1. React Native Documentation – <https://reactnative.dev/>
2. React Navigation Documentation – <https://reactnavigation.org/>
3. Android Studio Documentation – <https://developer.android.com/studio>